

Lecture 4 - When Complete Randomization is Infeasible

Matching and Weighting

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Outline

1 When complete randomization is infeasible

Examples

Identification under “selection on observable”

2 Various estimators under selection on observables

Subclassification

Matching

Weighting with propensity scores

3 Appendix

Complete randomization can be inefficient

Imagine you can only “afford” 20 consumers in your study. You know beforehand **gender** is an important factor for the study outcomes.

If you were leave it to chance to assign the 10 people:

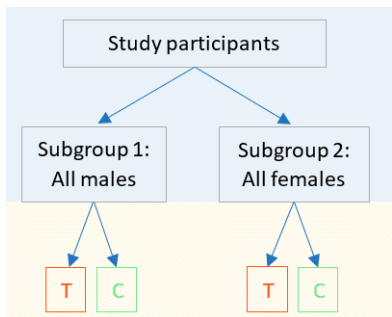
Randomly assign 10 consumers

```
people <- rep(c("female","male"),5)
unlist(rerun(10, sum(sample(people,10,replace = F)=="female")))
[1] 6 4 4 4 5 6 5 7 5 3
```

Blocking to increase efficiency

Instead, **we stratify the sample to females and males, and then randomize within each subgroup.**

- 1 Reduces sampling uncertainty and improves precision;
- 2 Ensures that certain subgroups are available for analysis (heterogeneity).



The treatment and control groups are forced to be equivalent regarding gender

Blocking is efficient: An example

store	block	All Stores		Block A		Block B	
		Y_1	Y_0	Y_1	Y_0	Y_1	Y_0
1	A	4	1	4	1		
2	A	5	3	5	3		
3	A	6	2	6	2		
4	A	4	1	4	1		
5	A	3	2	3	2		
6	A	2	1	2	1		
7	B	10	8			10	8
8	B	13	10			13	10
9	B	11	9			11	9
10	B	10	8			10	8
11	B	12	10			12	10
12	B	11	11			11	11
Mean		7.6	5.5	4.0	1.7	11.2	9.3
Variance		14.2	15.6	1.7	0.6	1.1	1.2

Under complete randomization:

$$\begin{cases} \text{ATE}_{\text{CR}} &= 7.6 - 5.5 = 2.1 \\ \text{SE}_{\text{CR}}^{\text{Neyman}} &= \sqrt{\frac{14.2}{12} + \frac{15.6}{12}} = \mathbf{1.57} \end{cases}$$

Blocking is efficient: An example

store	block	All Stores		Block A		Block B	
		Y_1	Y_0	Y_1	Y_0	Y_1	Y_0
1	A	4	1	4	1		
2	A	5	3	5	3		
3	A	6	2	6	2		
4	A	4	1	4	1		
5	A	3	2	3	2		
6	A	2	1	2	1		
7	B	10	8			10	8
8	B	13	10			13	10
9	B	11	9			11	9
10	B	10	8			10	8
11	B	12	10			12	10
12	B	11	11			11	11
Mean		7.6	5.5	4.0	1.7	11.2	9.3
Variance		14.2	15.6	1.7	0.6	1.1	1.2

Under block randomization:

$$\begin{aligned} \text{ATE}_{\text{BR}} &= \sum_{b=1}^B \frac{N_b}{N} \text{ATE}_b = \frac{6}{12} \cdot 2.3 + \frac{6}{12} \cdot 1.9 = 2.1 \\ \text{SE}_{\text{BR}}^{\text{Neyman}} &= \sqrt{\sum_{b=1}^B \left(\frac{N_b}{N} \text{SE}_b^{\text{Neyman}} \right)^2} = \mathbf{0.62} \end{aligned}$$

Why blocking increases efficiency

Proposition

Blocking on pre-treatment variables that predict potential outcomes weakly reduces the sampling variance of the treatment-effect estimator.

Let B denote blocks, $p_b = n_b/n$, and estimate

$$\hat{\tau}_{block} = \sum_b p_b (\bar{Y}_{1b} - \bar{Y}_{0b}).$$

With equal treatment probability q within each block:

$$Var(\hat{\tau}_{CR}) - Var(\hat{\tau}_{block}) = \frac{1}{n} \left[\frac{Var_b(\mu_{1b})}{q} + \frac{Var_b(\mu_{0b})}{1-q} - Var_b(\tau_b) \right].$$

- If blocks predict outcomes, $Var_b(\mu_{1b})$ and $Var_b(\mu_{0b})$ are large.
- If τ_b are not heterogeneous across blocks, $Var_b(\tau_b)$ is small.

$$\Rightarrow Var(\hat{\tau}_{block}) \leq Var(\hat{\tau}_{CR})$$

Key application: detecting small, marginal effects

Many experiments today are not testing large primary effects, but small marginal effects.

Psychology examples

Small effects often hidden by noisy outcomes



Subtle framing effects

Different descriptions, tiny attitude shifts



small shift



Emotion / attention primes

Brief cues shaping judgment or recall



small shift



Norm messages

Social information nudging behavior



small shift



Bias, trust, risk perception

Small interventions on latent evaluations



small shift

Consumer behavior examples

Tiny nudges can matter, but require precision



Sustainability / health labels

Small changes in product evaluation



small shift



Price-framing effects

Equivalent prices, different perceptions



small shift



Scarcity messages

Limited quantity or limited time cues



small shift



Ad-copy / recommendation nudges

Micro-variations in persuasive content



small shift

Key application: detecting small, marginal effects

For a two-arm experiment, approximate required sample per group:

$$n \approx \frac{16}{d^2}$$

for 80% power at $\alpha = .05$.

$d = .20 \Rightarrow n \approx 400$ per group, $d = .10 \Rightarrow n \approx 1600$ per group.

Blocking logic

Block on those variables before randomization. By reducing residual outcome variance, the same marginal effect can be detected with a smaller sample.

Block design as an algorithm

Steps in block design:

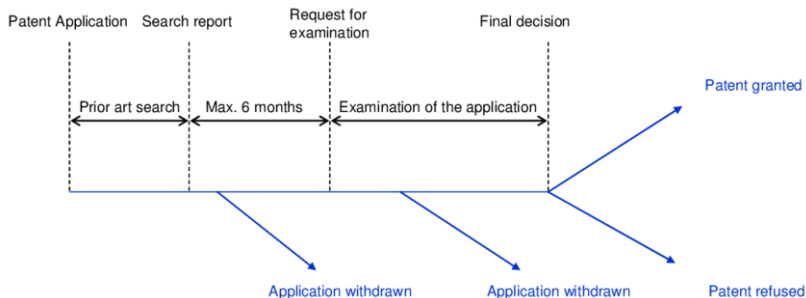
- 1 **Identify blocking variables** Select pre-treatment variables that theory or prior data suggest predict potential outcomes.
- 2 **Create blocking scheme** Define categories or strata.
- 3 **Pre-gather participant information** Measure these variables before treatment assignment.
- 4 **Form blocks** Group participants with similar values on the blocking variables.
- 5 **Randomize within blocks** Assign treatment and control separately inside each block.

Examples

Identification under “selection on observable”

3 Appendix

Patent examination



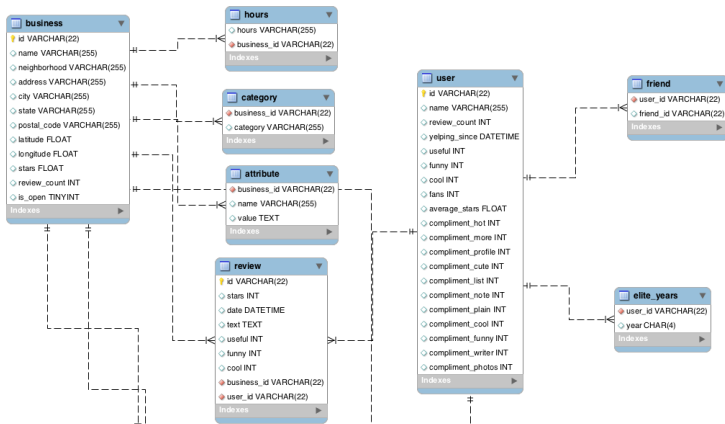
- **Treatment:** Patent granted.
- **Assignment:** Patent examination.

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Yelp Elite

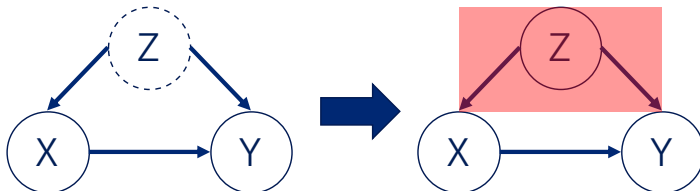


Almost all information is know, but not how Yelp selects Elites.

1

1

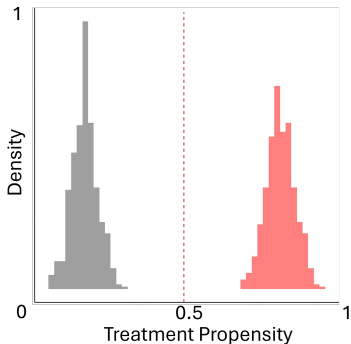
- ① $(Y_i^1, Y_i^0) \perp D_i \mid X_i$ (conditional unconfoundedness)
- ② $p_i(D_i = 1 \mid X_i) \in (0, 1)$ (common support)



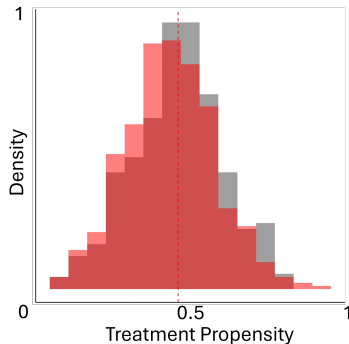
It's essentially a "conditioning" identification strategy

Understanding common support

The treatment propensity of the treated vs. control group must be overlapped.



(a) Non-overlapped



(b) Overlapped

$$\begin{aligned} E[Y_i^1 | X_i, D_i = 1] &= E[Y_i^1 | X_i] \\ E[Y_i^0 | X_i, D_i = 0] &= E[Y_i^0 | X_i] \end{aligned}$$

$$\tau_{\text{ATE}} = \int \left(E \left[Y_i^1 \mid X_i \right] - E \left[Y_i^0 \mid X_i \right] \right) dP(X_i)$$

Identification results: ATT

Under milder assumptions, in practices, also identify **ATT** (average treatment effects on the treated units).

$$ATT = E[Y_i^1 - Y_i^0 \mid D_i = 1]$$

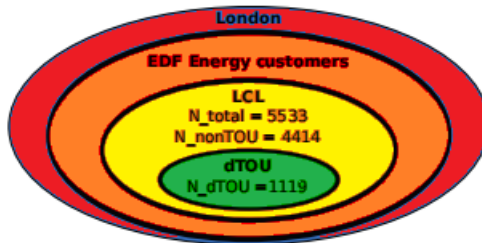
If we assume a milder version of unconfoundedness: $Y_i^0 \perp D_i \mid X_i$ and common support $p_i(D_i = 1 \mid X_i) < 1$, ATT is identified.

$$E[Y_i^0 | X_i, D_i = 1] = E[Y_i^0 | X_i]$$

$$E[Y_i^1 | X_i, D_i = 1], \text{ directly from data}$$

An example of ATT: “Low Carbon London” trial

Figure 1: Venn diagram illustrating the sample selection in LCL trial



The control group is randomly drawn and therefore ATT is identified.

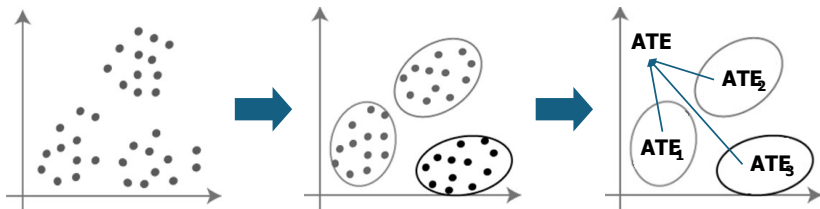
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- ② Various estimators under selection on observables
 - Subclassification
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Subclassification: the process

Given $(Y_i^1, Y_i^0) \perp D_i \mid X_i$, emulate blocking by creating “blocks” based on X_i :

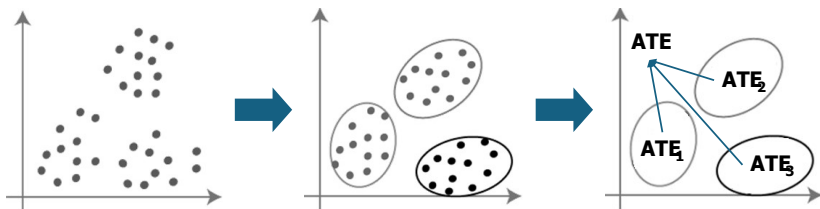
- Discrete variables: to use directly.
- Continuous variables: to first discretize and then use.



Subclassification: the process

By creating K groups, with size N_k for a group k :

$$\begin{cases} \hat{\tau}_{\text{ATE}} &= \sum_{k=1}^K \frac{N_k}{N} (\bar{Y}_1^k - \bar{Y}_0^k) \\ \hat{\tau}_{\text{ATT}} &= \sum_{k=1}^K \frac{N_k^1}{N^1} (\bar{Y}_1^k - \bar{Y}_0^k) \end{cases}$$



Subclassification: an example

Usage X_i	Average Purchase		Diff.	# Ad Viewers N_k^1	# Units N_k
	With Ads $\bar{Y}_1^k$	No Ads $\bar{Y}_0^k$			
Heavy	€28	€24	€4	3	10
Light	€22	€16	€6	7	10
Total				10	20

$$\hat{\tau}_{\text{ATE}} = \sum_{k=1}^K \left(\bar{Y}_1^k - \bar{Y}_0^k \right) \frac{N_k}{N} = 4 \cdot \frac{10}{20} + 6 \cdot \frac{10}{20} = \text{€}5$$

$$\hat{\tau}_{\text{ATT}} = \sum_{k=1}^K \left(\bar{Y}_1^k - \bar{Y}_0^k \right) \frac{N_k^1}{N^1} = 4 \cdot \frac{3}{10} + 6 \cdot \frac{7}{10} = \text{€}5.4$$

Subclassification: potential problems

Usage	Average Purchase		Diff.	# Ad Viewers	# Units
	With Ads	No Ads			
Heavy, Male	€28	22	€4	3	7
Heavy, Female	??	€24	??	0	3
Light, Male	€21	€16	€5	3	4
Light, Female	€23	€17	€6	4	6
Total				10	20

$$\hat{\tau}_{ATE} = \sum_{k=1}^K \left(\bar{Y}_1^k - \bar{Y}_0^k \right) \frac{N_k}{N} = ???$$

$$\hat{\tau}_{ATT} = \sum_{k=1}^K \left(\bar{Y}_1^k - \bar{Y}_0^k \right) \frac{N_k^1}{N^1} = ???$$

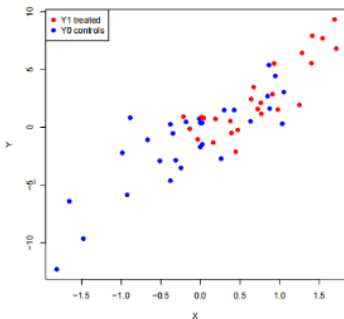
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Matching: the basic idea

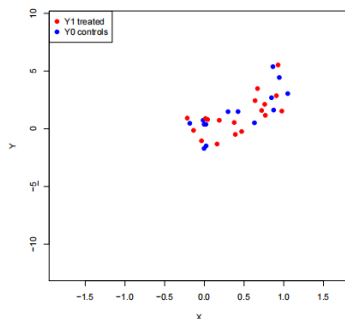
Recall the conditional uncoufoundedness:

$$E[Y_i^D | X_i, D_i] = E[Y_i^D | X_i]$$

Implication: **to equalize the distribution of X_i for the treated and control.**



(a) Before Matching



(b) After Matching

Matching: a general procedure



1. SELECT A MATCHING METHOD.



2. OBTAIN THE MATCHED SAMPLES.



3. EVALUATE THE MATCHING PERFORMANCES.



4. CALCULATE THE TREATMENT EFFECTS.

Matching: a simple example

Unit	Potential Outcome		Treatment	Variables
	Under Treatment	Under Control		
i	Y_i^1	Y_i^0	D_i	X_i
1	6	?	1	3
2	1	?	1	1
3	0	?	1	10
4		0	0	2
5		9	0	3
6		1	0	-2
7		1	0	-4

What is $\hat{\tau}_{ATT} = \frac{1}{N_1} \sum_{i:D_i=1} (Y_i^1 - \overline{Y_j^0})$?

Match and plug in potential outcomes.

Matching: a simple example

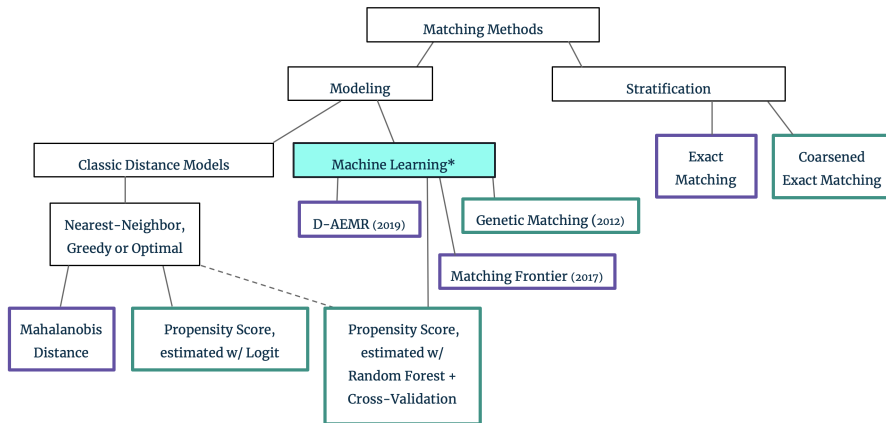
Unit	Potential Outcome		Treatment	Variables
	Under Treatment	Under Control		
i	Y_i^1	Y_i^0	D_i	X_i
1	6	9	1	3
2	1	0	1	1
3	0	9	1	10
4		0	0	2
5		9	0	3
6		1	0	-2
7		1	0	-4

What is $\hat{\tau}_{ATT} = \frac{1}{N_1} \sum_{i:D_i=1} (Y_i^1 - \overline{Y_j^0})^1$?

$$\hat{\tau}_{ATT} = \frac{1}{3} ((6 - 9) + (0 - 1) + (9 - 1)) = -3.7$$

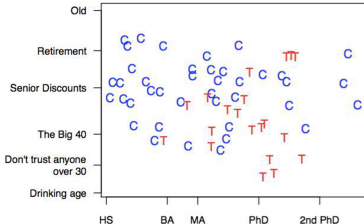
¹Note: ATC (ATE on the control units) can be calculated similarly.

A classification of matching methods

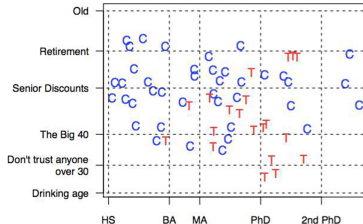


Stratification: coarsened exact matching

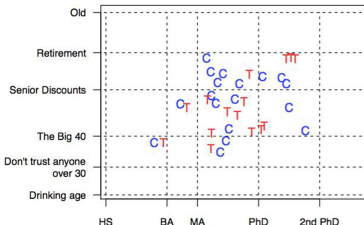
Step 1: to check overlaps



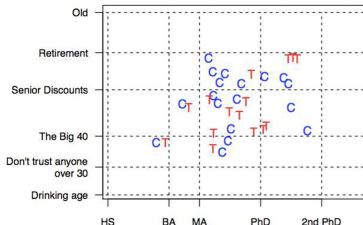
Step 2: to create bins



Step 3: to prune control samples



Step 4: to calculate treatment effects



Modeling: classic distance models

One way is to define “closeness” by a distance metric.

Classic distance metrics

Suppose we have a treated unit i and a control unit j .

A set of continuous characteristics for i and j , X_i and X_j .

Two most frequently used distance metrics:

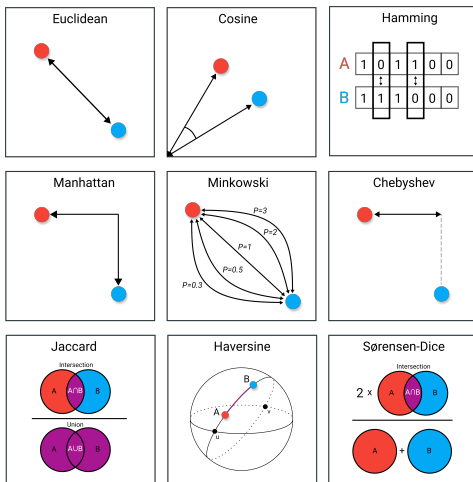
$$\text{Mahalanobis distance: } \sqrt{(X_i - X_j)' \Sigma^{-1} (X_i - X_j)}$$

$$\text{Euclidean distance: } \sqrt{(X_i - X_j)' (X_i - X_j)}$$

For an “exact match”, the distance metrics are zeros.

Modeling: classic distance models

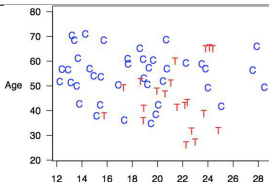
A rich selection of distance metrics (probably too many...)



Examples of Distance Metrics

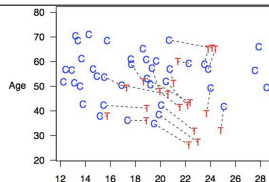
Modeling: classic distance models

Step 1: to calculate the distances

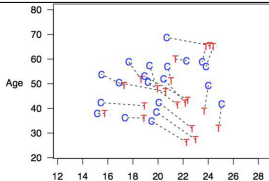


Step 2: to find the matches

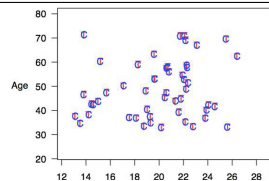
Methods: greedy vs. optimal



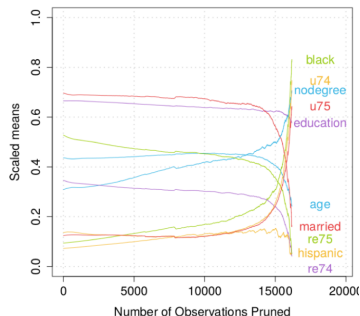
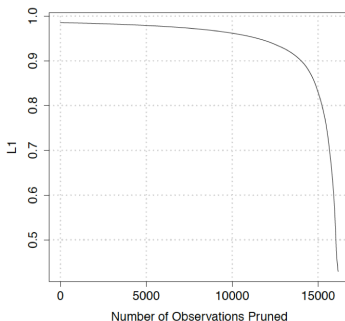
Step 3: to prune the unmatched control units



Step 4: to obtain the matched samples



Modeling: matching frontier



Source: King et al. (2017)

Formalize a standard bias-variance trade-off in matching:

Poor covariate balance can lead to biased causal effect estimates but pruning too many observations can increase the variance of the estimates.

Evaluate matching: balance tests

The performance of matching depends on **the resulting balance of X_i between matched samples²**.

How should one assess the balance of matched data?

²See the appendix for a formal proof.

Evaluate matching: balance tests

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How should one assess the balance of matched data?

- Ideally, compare the joint distribution of all X_i for the matched samples.

²See the appendix for a formal proof.

Evaluate matching: balance tests

The performance of matching depends on **the resulting balance of X_i between matched samples**².

How should one assess the balance of matched data?

- Ideally, compare the joint distribution of all X_i for the matched samples.
- In practice, this is impossible when X_i is high-dimensional and so check various lower-dimensional summaries.

²See the appendix for a formal proof.

Evaluate matching: balance tests

The performance of matching depends on **the resulting balance of X_i between matched samples²**.

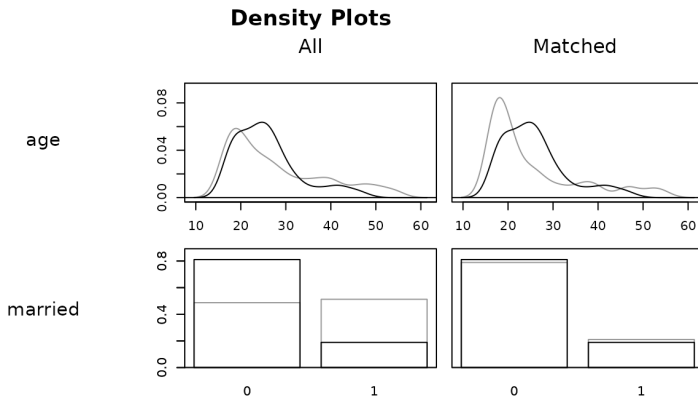
How should one assess the balance of matched data?

- Ideally, compare the joint distribution of all X_i for the matched samples.
- In practice, this is impossible when X_i is high-dimensional and so check various lower-dimensional summaries.
- Statistical tests are often used; plotting is also common in practice.

²See the appendix for a formal proof.

Evaluate matching: balance tests

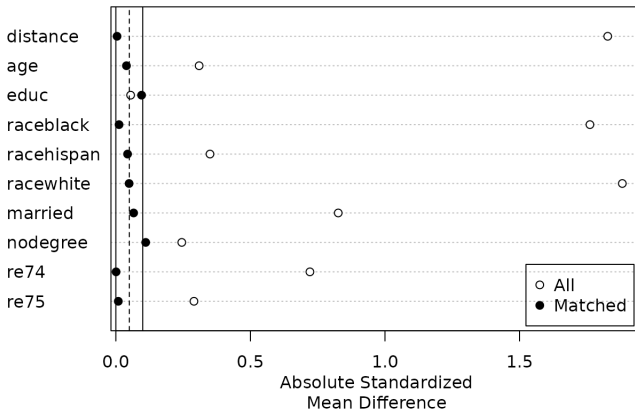
The performance of matching depends on **the resulting balance of X_i between matched samples**³.



³See the appendix for a formal proof.

Evaluate matching: balance tests

The performance of matching depends on **the resulting balance of X_i between matched samples**⁴.



⁴See the appendix for a formal proof.

Balance tests fallacy

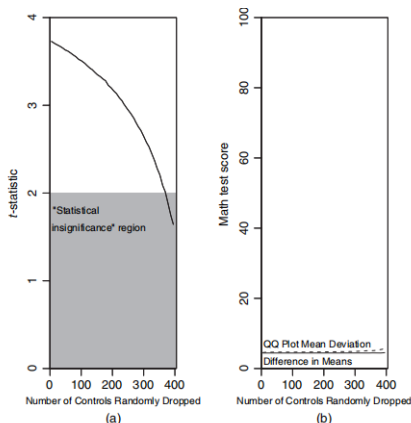


Fig. 1. Dangers in relying on t -statistics as a measure of balance (average value of a measure of balance when a given number of control units are randomly dropped from the data set (out of a total of 434)): with larger numbers of control units dropped (i.e. smaller numbers of control units in the resulting sample), the value of the t -statistic becomes closer to 0, falsely indicating improvements in balance, even though true balance does not vary systematically across the data sets (and efficiency declines); the difference in means and quantile-quantile plot mean deviation, which are given in (b), correctly indicate no change in bias as observations are randomly dropped

Source: Imai et al. (2008)

Matching: potential issues

Some difficult decisions in distance matching

- 1 With replacement or without replacement?
- 2 Which distance metrics to use?
- 3 A single closest control unit or multiple closer control units?

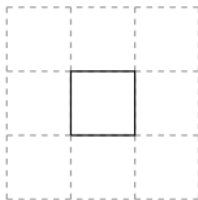
In practice, you almost always need to use multiple matching methods to show the “**robustness**”.

Matching: potential issues

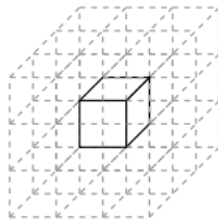
Difficult to find “good matches” with many continuous variables or variables with large variance...



(a)



(b)



(c)

The probability of the “focal” region decreases exponentially:
 $1/3 \rightarrow 1/9 \rightarrow 1/27$.

Source: Betancourt (2017)

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Propensity score to the rescue

Instead of working with X_i , why not **a statistic** of X_i ?

Propensity Score: $e(X_i) = P(D_i = 1 | X_i)$

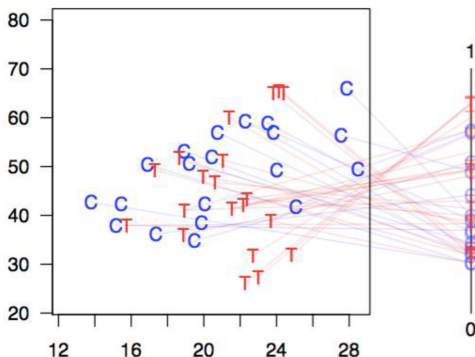
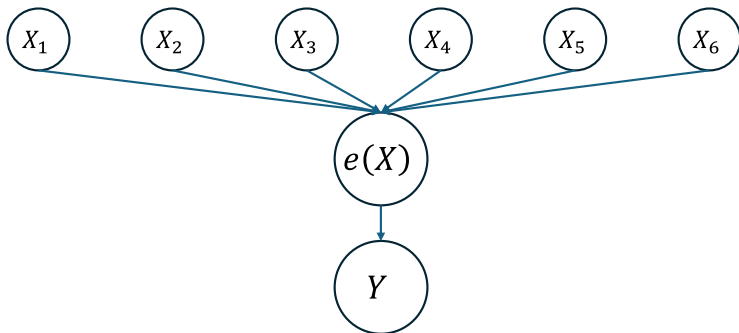


Figure 2: Propensity score “shrinks” X_i into a single index.

Identification results (Proof on slide 67)

Given **conditional unconfoundedness** and **common support**,

- 1 The propensity score is a sufficient statistic of X_i , such that $D_i \perp X_i \mid e(X_i)$.
- 2 The treatment is unconfounded given the propensity score: $D_i \perp Y_i^1, Y_i^0 \mid e(X_i)$.



Implications of the identification results

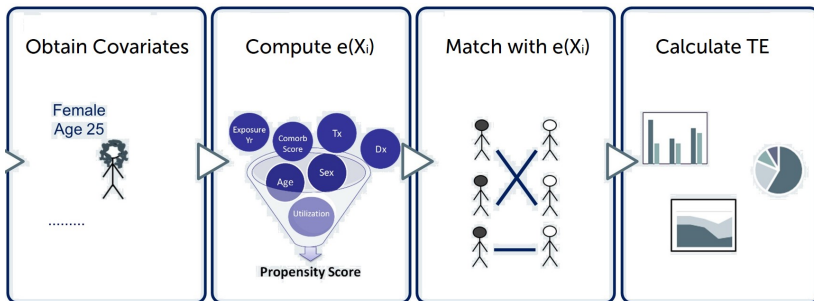
Adjustment of the treatment-control differences in $e(X_i)$ is sufficient to remove all biases from the differences in X_i .

$$\begin{aligned} E[Y_i^1 - Y_i^0 \mid D_i, e(X_i)] &\stackrel{\text{Unconfounded}}{=} E[Y_i^1 - Y_i^0 \mid e(X_i)] \\ &\stackrel{\text{Definition}}{=} E[Y_i^1 - Y_i^0 \mid P(D_i = 1 \mid X_i)] \end{aligned}$$

Suggests a two step procedure to estimate treatment effects:

- 1 **Estimate the propensity score** $\hat{e}(X_i)$.
- 2 **Estimate the expectations** of potential outcomes conditional on $\hat{e}(X_i)$.

Matching with propensity score



Illustrating propensity score matching

Weighting with propensity scores

Matching throws away samples to gain balance.
The “efficiency” is sacrificed for “precision.”

Question: Can we do better?

Observe two equations given propensity scores⁵:

$$\begin{cases} E \left(\frac{D_i \cdot Y_i^{\text{obs}}}{e(X_i)} \right) &= E [Y_i^1 | X_i] \\ E \left(\frac{(1-D_i) \cdot Y_i^{\text{obs}}}{1-e(X_i)} \right) &= E [Y_i^0 | X_i] \end{cases}$$

⁵The original construction is in Horvitz and Thompson (1952).

Weighting with propensity scores

Technical Proof:

$$\begin{aligned}
 E\left(\frac{D_i \cdot Y_i^{\text{obs}}}{e(X_i)}\right) &= E\left(\frac{D_i \cdot Y_i^1}{e(X_i)}\right) \\
 &= E\left[E\left(\frac{(D_i = 1) \cdot Y_i^1}{e(X_i)} \middle| X_i\right)\right] \\
 &= \frac{E_D[D_i = 1 \mid X_i] E[Y_i^1 \mid X_i]}{e(X_i)} \\
 &= E[Y_i^1 \mid X_i]
 \end{aligned}$$

Similar proof holds for the second equality.

The weighting estimator

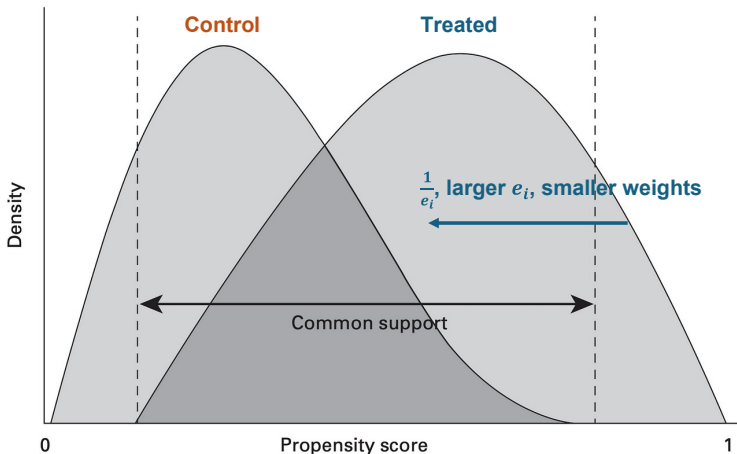
Based on the equations, we can construct two estimators:

$$E[\widehat{Y_i^1} | X_i] = \frac{1}{N} \sum_{i=1}^N \frac{D_i \cdot Y_i^{\text{obs}}}{e(X_i)}$$
$$E[\widehat{Y_i^0} | X_i] = \frac{1}{N} \sum_{i=1}^N \frac{(1 - D_i) \cdot Y_i^{\text{obs}}}{1 - e(X_i)}$$

The weighting estimator of ATE is defined as,

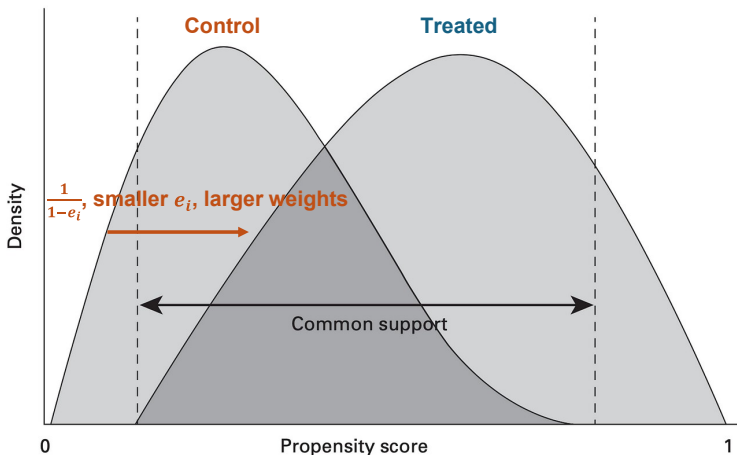
$$\begin{aligned}\hat{\tau} &= E[\widehat{Y_i^1} | X_i] - E[\widehat{Y_i^0} | X_i] \\ &= \frac{1}{N} \sum_{i=1}^N \left(\frac{D_i \cdot Y_i^{\text{obs}}}{e(X_i)} - \frac{(1 - D_i) \cdot Y_i^{\text{obs}}}{1 - e(X_i)} \right)\end{aligned}$$

The weights w_i is $\frac{1}{e(X_i)}$ for the treated and $\frac{1}{1-e(X_i)}$ for the control.



“Expanding” the common support

The weights w_i is $\frac{1}{e(X_i)}$ for the treated and $\frac{1}{1-e(X_i)}$ for the control.



“Expanding” the common support

The weighted regression estimation

The weighted regression procedure

- 1 Estimate the propensity scores with $P(D_i = 1 \mid X_i)$, possibly with different methods and specifications.
- 2 Plug in estimated propensity score $\hat{e}(X_i)$ into a weighted regression.

$$Y_i = \alpha + \tau D_i + X_i \beta + \varepsilon_i \text{ with } w_i = \begin{cases} \frac{1}{\hat{e}(X_i)} & \text{if } D_i = 1 \\ \frac{1}{1 - \hat{e}(X_i)} & \text{if } D_i = 0 \end{cases}$$

The weighted regression for ATT and ATC

Some procedure, but with different weights:

- The weights (odds ratio) for ATT estimation is:

$$w_i = \begin{cases} 1 & \text{if } D_i = 1 \\ \frac{\hat{e}(X_i)}{1 - \hat{e}(X_i)} & \text{if } D_i = 0 \end{cases}$$

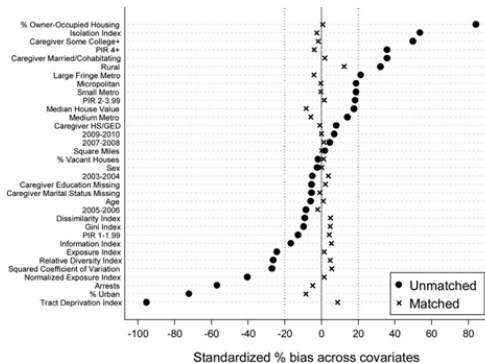
- The weights for ATC estimation is similar:

$$w_i = \begin{cases} \frac{1 - \hat{e}(X_i)}{\hat{e}(X_i)} & \text{if } D_i = 1 \\ 1 & \text{if } D_i = 0 \end{cases}$$

What is a good estimation of propensity score?

Balancing condition: samples after weights should show similar distributions of X_i

$$E \left[\frac{D_i \cdot X_i}{e(X_i)} - \frac{(1 - D_i) \cdot X_i}{1 - e(X_i)} \right] = 0$$



A set of small navigation icons typically found in Beamer presentations, including symbols for back, forward, search, and other slide controls.

Other ML methods in matching: see references

Genetic matching:

- Diamond, A., & Sekhon, J. S. (2013)

Dynamic almost exact matching with replacement:

- Dieng et al. (2019)

Propensity score matching with random forest:

- Krief and Diaz-Ordaz (2019), Ferri-Garcia and Rueda (2020)

Proof: matching is about covariate balancing

The conditional ATE or CATE is: $\tau(X_i) = E[Y_i^1 - Y_i^0 | X_i]$.

Under the assumptions of conditionally unconfounded, individualistic and probabilistic assignment, the variance for CATE is⁶,

$$\text{Var}(\tau(X_i)) = E \left[\frac{\sigma_c^2(X_i)}{1 - e(X_i)} + \frac{\sigma_t^2(X_i)}{e(X_i)} + (\tau(X_i) - \tau)^2 \right]$$

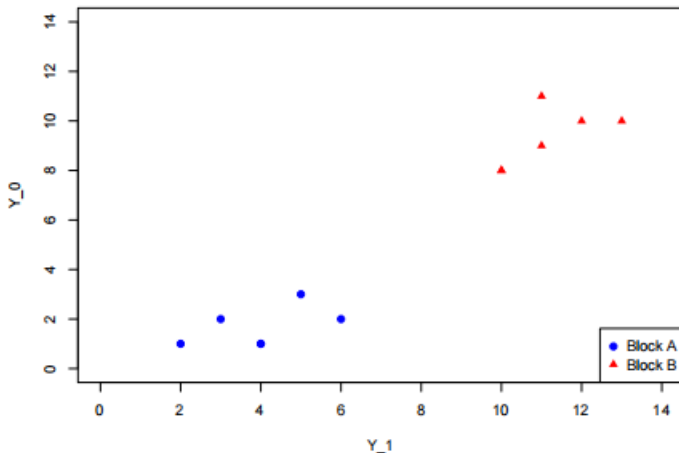
$e(X_i)$ the propensity to be treated, and τ the unconditional ATE.

The third term $(\tau(X_i) - \tau)^2$ goes to zero if the treatment effect is constant for the treated and the control.

This requires X_i has the same distributions in both groups, i.e., X_i is balanced.

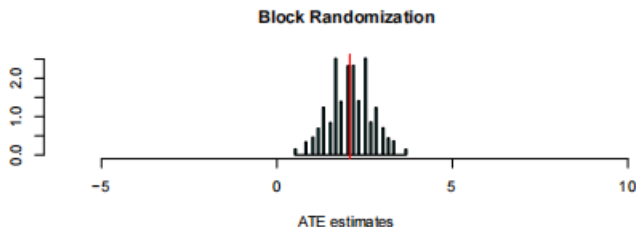
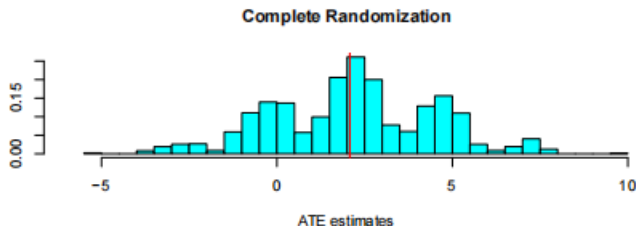
⁶Please see Imbens and Rubin (2015), p. 269 for the derivation. 

Blocking is efficient: Intuition



Potential outcomes across blocks

Blocking is efficient: Intuition



Sampling distribution of ATE estimates

Blocking: summary

What to block on?

- **“Block what you can/need, randomize what you can’t or don’t need.”**
- Variables linked to potential outcomes.
- Variables desired for subgroup analysis.

How to block?

- Stratification
- Pairwise assignment...

Proof: the sufficiency of propensity score

Here, we prove that the propensity score $e(X_i)$ is a sufficient statistic of X_i or:

$$D_i \perp X_i \mid e(X_i) \text{ or } P(D_i = 1 \mid X_i, e(X_i)) = P(D_i = 1 \mid e(X_i))$$

First, for the left-hand side:

$$P(D_i = 1 \mid X_i, e(X_i)) = P(D_i = 1 \mid X_i) = e(X_i)$$

Second, for the right-hand side:

$$\begin{aligned} P(D_i = 1 \mid e(X_i)) &= E[D_i \mid e(X_i)] \\ &= E[E[D_i \mid X_i, e(X_i)] \mid e(X_i)] \\ &= E[e(X_i) \mid e(X_i)] = e(X_i) \end{aligned}$$

The second equality is the iterated expectations and the third equality is proved above.

Proof: the conditional unconfoundedness of the propensity score

We want to prove that $D_i \perp Y_i^0, Y_i^1 \mid e(X_i)$, or

$$P(D_i = 1 \mid Y_i^0, Y_i^1, e(X_i)) = P(D_i = 1 \mid e(X_i))$$

For the left-hand side, we have:

$$\begin{aligned} P(D_i = 1 \mid Y_i^0, Y_i^1, e(X_i)) &= E[D_i \mid Y_i^0, Y_i^1, e(X_i)] \\ &= E[E[D_i \mid Y_i^0, Y_i^1, X_i, e(X_i)] \mid Y_i^0, Y_i^1, e(X_i)] \end{aligned}$$

By conditional unconfoundedness, $E[D_i \mid Y_i^0, Y_i^1, X_i, e(X_i)] = E[D_i \mid X_i, e(X_i)]$.

By the sufficiency of the propensity score, $E[D_i \mid X_i, e(X_i)] = E[D_i \mid e(X_i)]$. If we submit it to the equation above, we have:

$$E[E[D_i \mid e(X_i)] \mid Y_i^0, Y_i^1, e(X_i)] = E[D_i \mid e(X_i)] = P(D_i = 1 \mid e(X_i))$$

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